MIL-P-10096E
7 June 1985
SUPERSEDING
MIL-P-10096D
5 August 1977

MILITARY SPECIFICATION

PRINTER, CONTACT, PHOTOGRAPHIC PH-649( )/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of semi-automatic, motor-driven, floor-type, photographic contact printer designated as Printer, Contact, Photographic PH-649( )/U (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-F-336	- Film, Photographic, Photomechanical Black-And-White.
W-C-596/13C	- Connector, Plug, Electrical (Power, Three Wire, Polarized, Fixed Grounding Blade), Type UP221M.
CC-M-1807	- Motor, Alternating Current, Fractional And Integral Horsepower (500 HP And Smaller).

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-C-3432	- Cable And Wire, Electrical (Power And Control, Flexible And Extra Flexible, 300 And 600 Volts).
MIL-C-3885	- Cable, Assemblies And Cord Assemblies, Electrical.
MIL-P-4672	- Paper, Photographic, Black And White, Sensitized.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, NJ 07703-5304 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-4827	- Developer, Photographic Paper, Armed Forces Developer No. 25.
MIL-M-13231	- Marking Of Electronic Items.
MIL-F-14072	- Finishes For Ground Electronic Equipment.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-252	- Wired Equipment, Classification Of Visual And Mechanical Defects.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-810	- Environmental Test Methods.

2.1.2 Other Government documents. The following Government document form a part of this specification to the extent specified herein.

SPI 1G00478 - Special Packaging Instructions.

(Copies of specifications, standards and drawings required by manufacturers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
ANSI PH3.8 - Contact Printers, Specification For.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein, the test of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The printer is a floor model, semi-automatic, motor driven device used for mass production of contact prints from negatives up to 11 by 14 inches (279 by 356mm) in size. It is capable of either single or continuous cycle operation.

3.2 First article. The quantity of samples specified in the contract shall be subjected to first article inspection (see 4.3 and 6.2b).

3.3 Material. The material shall conform to requirement 18 of MIL-STD-454 and as specified herein. However, when a definite material is not specified, a material shall be used which will enable the printer to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled, virgin and reclaimed material. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.4 Design.

3.4.1 Printer body. The printer body shall be sturdy, metal construction of such proportions to house the printing lamps, timer, driving motor and mechanism, printing glasses and plates. There shall be no leakage of white light. Suitable light-trap slots may be provided, if needed, to meet this requirement.

3.4.2 Diffusing screen. An easily accessible, removable and adjustable, fine ground or opal, glasss diffusion screen shall be provided between the printing lamps and the printing surface (glass) to provide even distribution of illumination.

3.4.3 Printing glass. The printing glass shall be not less than .025 inch (6.4mm) thick plate glass, readily removable for cleaning. It shall be free of bubbles, striae, scratches and any defect having an adverse effect on uniformity of illumination.

3.4.4 Negative glass. The negative glass shall be a mud ground plate glass not less than 0.125 inch (3.2mm) thick. It shall be positioned between the diffusion screen and printing glass and shall be readily removable for masking and cleaning. It shall be free of bubbles, scratches, striae and any defect having an adverse effect on uniformity of illuminations.

3.4.5 Masking frame assembly. Readily removable masking blades, mounted on a masking frame, shall be constructed as to prevent damage to the photographic paper or film when either or both are moved under the mask. The sleeves by which the blades are attached to the frame shall be so constructed that they can be positioned along the frame without binding or chattering. Angles between adjacent masking blades shall be 90° +10 minutes of arc. The blades shall be capable of being accurately aligned (either parallel or at right angles, or both) whether clamped or unclamped. The blades shall be constructed of austenitic stainless steel and shall be resistant to deformation.

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3.4.6 Scales and guide lines. One masking blade along the longer dimension and one masking blade along the shorter dimension of the printer shall be graduated (marked) in inches and centimeters with intermediate graduations (markings) at 0.25 inch and 0.5 centimeter intervals starting at zero in the center of the photo printing aperture and extending in both directions for the full aperture. The graduations shall be on the blade edges nearest to the printing aperture. The other two masking blades shall be provided with parallel guide lines spaced and marked at 0.25 inch and 0.5 centimeter intervals. The margin guide lines shall permit a white border of not less than 3 inches (76mm).

3.4.7 Platen. The platen shall be covered with a high quality sponge rubber or felt padding, protected with a suitable covering, and shall provide contact sufficient to permit the equipment to meet the operational requirements of this specification.

3.4.8 Counter. The printer shall include a counter of not less than three digits which registers the number of prints which have been made in increments of 1, from 000 to 999. The counter shall be quickly manually resettable to zero.

3.4.9 Light leakage and protection. The printer shall be so constructed as to minimize light leakage and to provide protection, such as by insulation of the lamp compartment door and switch panel from the main body of the printer, against accidental contact with potentials greater than 30 volts by operating personnel.

3.5 Electrical requirements.

3.5.1 Safety of electrical components. All electrical components which are not otherwise described in this specification shall be Underwriters' Laboratories approved. All wiring and electrical contacts, with the exception of the power and foot switch cords and the foot switch, shall be inclosed so that the operator cannot accidentally touch any of them during normal operation of the equipment. Each door, plate and cover of any type which covers uninsulated electrical contact points must contain a caution notice to indicate to maintenance personnel that exposed contact points are inclosed; each notice shall also include the approximate voltage(s) of the inclosed uninsulated contact points.

3.5.2 Power requirements. The printer shall operate satisfactorily from a 115 $\pm$10 volt, single phase, 60 hertz (Hz) alternating current (ac) power source.

3.5.3 Power cable. The printer shall be provided with a flexible cable type CO-03-MGF (3/16) 0360 conforming to MIL-C-3432 (see 6.7) and color coded black, white, and green; 144 (+1 or -0) inches (3658 (+25 or -0)mm) in length; one end terminated at the printer and the other end terminated in connector plug UP-221M conforming to W-C-596/13. The green coded conductor in the cable shall provide an electrical power ground for the equipment. Assembly of the power cable shall conform to MIL-C-3885.

3.5.4 Wiring. Wiring shall be of heat-resistant (asbestos covered or equivalent heat-resistant) insulation. Wire size shall conform to operational load. Power input shall be fused consistent with wire size and load. Each wire shall be color coded or otherwise identified (marked) in association with its particular circuit as an aid in circuit tracing. Wires longer than 6 inches (152mm) shall be marked at each end if color coding is not over the entire length.

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3.5.5 Motor. The printer shall contain a 1/6 horsepower (124 watt), or greater, 1725 rpm synchronous motor conforming to CC-M-1807.

3.5.5.1 Motor drive mechanism. The drive mechanism shall be actuated either by manually starting the platen on its downward travel (stroke) or by means of a foot switch. Provision shall be made so that the drive mechanism can be halted at any point in the operation cycle.

3.5.6 Switches. The printer shall be equipped with a power on-off switch, "master" printing lamp switch, and individual printing lamp toggle switches, all conforming to Underwriters' Laboratories requirements.

3.5.7 Safe and dial lamps. Each printer shall be equipped with one 10 watt, 120 volt, orange/amber, medium screw base safe lamp and one 7.5 watt, red glass, extended life, 3000-hour type candelabra screw base dial lamp (for the timer dial), or equally suitable items.

3.5.8 Printing lamps. The printer shall be provided with 12 white glass, 25 watt, 120 volt lamps or an equivalent light source which shall provide the uniformity of illumination specified herein.

3.5.9 Timing device. The printer shall be provided with an integral timing device which together with the master lamp switch and drive actuator shall control printing lamps. The device (timer) shall be a calibrated synchronous motor type, having a range not less than 0.75 through 15 seconds. The values at each setting shall be consistent within 2 percent of the calibrated time. Each calibration shall be accurate to $\pm$0.25 seconds for all settings of 5 seconds or less, and $\pm$0.50 seconds for all settings above 5 seconds. The timer dial shall be illuminated by a safe light controlled by an on-off switch. A switch activated by the closing of the platen shall start the timer. Privision shall be made for lengthening the exposure after the timing cycle has started. Means shall be provided so that the timer may be cut out of the circuit and exposure made by means of a manual printing lamp switch. The timer shall be readily removable for servicing.

3.5.10 Leakage current. The leakage current between adjacent conductors or between any conductor and power ground shall be not more than 2 milliamperes.

3.5.11 Dielectric strength. When tested as specified in 4.6.1.4, there shall be no voltage flash-over or breakdown.

3.6 Durability. When tested as specified in 4.6.2, the printer shall be operated for not less than 2500 cycles without failure. A burned out lamp shall not be considered a failure.

3.7 Environmental. The printer shall withstand individual environmental conditions specified herein, or combinations thereof, without damage, performance degradation, or malfunction when tested as specified in 4.6.3.

3.7.1 Operational.

a. Temperature 77 $\pm$9°F (25 $\pm$5°C).
b. Humidity: Up to 95 percent relative humidity.

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3.7.2 Storage.

a. Temperature: -60 ±5°F (-51 ±3°C) for a period of 72 hours.

b. Humidity: Up to 95 percent relative humidity.

3.8 Performance Requirements.

3.8.1 Intensity of illumination. The intensity of illumination at the exposure plane of the printer shall be such as to produce, in 4-6 seconds, a photographic print having a reflection density of 1.00 ±0.05 when all printing lamps are in use and tested as specified in 4.6.4.4.

3.8.2 Uniformity of illumination. The illumination at any corner of the exposure plane shall not vary more than 10 percent from the average illumination at all 4 corners. The ratio of average corner to center illumination shall be not less than 65 percent, and the entire field shall be free from local irregularities having intensity differences of more than 15 percent per inch (25.4mm).

3.8.3 Uniformity of contact. There shall be no region in the body of the print of more than 0.375 inch (9.5mm) which departs from a uniform grey tone and the requirements of ANSI PH3.8 when tested as specified in 4.6.4.6.

3.9 Electromagnetic interference. The printer shall meet the electromagnetic interference requirement specified in MIL-STD-461, table 1-11, class C3. If the contractor considers it desirable to modify the equipment in order to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.10 Dimensions and weight. The overall dimensions of the printer shall be approximately 26 inches (660mm) wide, 30 inches (762mm) deep and 58 inches (1473mm) high. The weight shall not exceed 200 pounds (90.7kg).

3.11 Interchangeability. Like units, assemblies, subassemblies and replacement parts shall conform to requirement 7 of MIL-STD-454.

3.12 Finish, protective. The printer shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type I surfaces shall be final film E, semigloss, light-gray enamel, conforming to MIL-F-14072.

3.13 Marking.

3.13.1 General. Each printer shall be provided with a nameplate in accordance with MIL-M-13231.

3.13.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the printer.

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3.13.3 Serial numbers. Each printer shall be serial numbered.

3.14 Workmanship. The printer shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following requirements specified herein:

3.4.1 Printer body.
3.4.2 Diffusing screen.
3.4.3 Printing glass thickness.
3.4.4 Negative glass thickness.
3.4.5 Masking blade angle.
3.4.6 Scales and guide lines.
3.4.8 Counter.
3.4.9 Light leakage.
3.5.1 Safety of electrical components.
3.5.3 Power cable length.
3.5.4 Wiring.
3.5.5 Motor.
3.5.6 Switches.
3.5.7 Safe and dial lamps.
3.5.8 Printing lamps.
3.10 Dimensions and weight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspection covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.6.8).

4.3 First article.

4.3.1 First article units. When specified in the contract, the contractor shall furnish first article units of the complete printer.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for group A, group B and group C. The inspections shall be performed in the following order: 4.4, group A and group B for all units, and group C as specified in table III. After completion of group C tests, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements.

Item	Where required
Cable assembly	3.5.3
Finish	3.12
Marking	3.13

4.5 Quality conformance inspection. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit shall be subjected to the inspections specified in table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined, or as specified herein, shall be in accordance with MIL-STD-105, using the general inspection levels and acceptable quality levels (AQL) indicated in table I.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective) Major	Minor
Visual and mechanical	3.14	4.6.7	1.0	4.0
Electrical tests				
Power requirements	3.5.2	4.6.1.1	1.0 for combined group	– 1/
Timing device	3.5.9	4.6.1.2		
Operational test				
Intensity of illumination	3.8.1	4.6.4.4	1.0 for combined group	
Uniformity of illumination	3.8.2	4.6.4.5		
Uniformity of contact	3.8.3	4.6.4.6		

1/ Electrical and operational defects are all major.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table II.

TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective)
Interchangeability	3.11	4.6.6	6.5
Leakage current	3.5.10	4.6.1.3	4.0
Dielectric strength	3.5.11	4.6.1.4	6.5

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4.5.2.1 Group B sampling plan. Sampling for group B shall be in accordance with inspection level S-4 of MIL-STD-105. AQL shall be as specified in table II.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table III.

Inspection	Requirement paragraph	Test paragraph
Electromagnetic interference	3.9	4.6.5
Low temperature	3.7.1, 3.7.2	4.6.3.1.1
High temperature	3.7.1, 3.7.2	4.6.3.1.2
Humidity test	3.7.1, 3.7.2	4.6.3.1.3
Durability	3.6	4.6.2

4.5.3.1 Sampling for group C inspection. One sample unit shall be selected at random from every 100 units, or fraction thereof, produced. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed as specified in table III.

4.5.3.3 Group C failures. Action required relative to group C failure shall be as specified in the contract (see 6.2).

4.5.4 Reinspection of group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Methods of inspection.

4.6.1 Electrical tests.

4.6.1.1 Power requirements. The printer shall be subjected to the tests of 4.6.4.4, 4.6.4.5, and 4.6.4.6 under the following two line-input conditions:

a. 105 (+1 or -0) volts at 60 Hz, single phase.
b. 125 (+0 or -1) volts at 60 Hz, single phase.

Nonconformance to 3.5.2 shall constitute failure of this test.

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4.6.1.2 Timing device. The timing device (timer) shall be tested with an electric stop clock which is accurate to ±0.02 second. A minimum of 10 settings shall be tested and at least 10 values taken at each setting. The settings chosen shall vary over the entire range of the timer. At least 5 of the chosen settings shall be below 7 seconds. Consistency of values and accuracy of calibration shall be recorded. Nonconformance to 3.5.9 shall constitute failure of this test.

4.6.1.3 Leakage current. Leakage current shall be determined with the printer operating from a 115 ±10 volts power source and with all lamps removed from their sockets. Nonconformance to 3.5.10 shall constitute failure of this test.

4.6.1.4 Dielectric strength. The insulation (break-down) strength of all electrical circuits shall be tested to withstand twice the rated voltage plus 1000 volts ac or dc. The test potential shall be applied for one minute between adjacent conductors and between all conductors and power ground. Nonconformance to 3.5.11 shall constitute failure of this test.

4.6.2 Durability. The equipment shall be operated for a total of 2500 cycles at intervals of 5-7 seconds each. Upon completion, examine the equipment and subject it to the tests of 4.6.4.4, 4.6.4.5 and 4.6.4.6. Nonconformance to 3.6 shall constitute failure of this test.

4.6.3 Environmental.

4.6.3.1 Temperature.

4.6.3.1.1 Low temperature. The printer shall be tested in accordance with method 502.1, procedure I of MIL-STD-810. The storage temperature shall be -60 ±5°F (-51 ±3°C) for 72 hours, and the operating temperature shall be 77 ±9°F (25 ±5°C).

4.6.3.1.2 High temperature. The printer shall be tested in accordance with method 501.1, procedure I of MIL-STD-810. For step 4, the internal chamber temperature shall be adjusted to 77 ±9°F (25 ±5°C).

4.6.3.1.3 Humidity (moisture-resistance). The printer shall be tested in accordance with method 507.1, procedure II, of MIL-STD-810, with test measurements during step 4 and the last 5-hour period of the 48-hour cycle in step 6. Step 6 shall be limited to one cycle. Nonconformance to 3.7 shall constitute failure of these tests.

4.6.4 Performance.

4.6.4.1 Test conditions. Unless otherwise specified herein, such as in temperature and humidity tests, the following tests shall be conducted at ambient temperature 77 ±9°F (25 ±5°C) and relative humidity of 25 to 50 percent.

4.6.4.2 Test negative. A test negative shall be made from an engraver's 120 lines/inch (5 lines/mm) screen in such a manner as to have alternating clear and opaque regions of equal area. The base material of the test negative shall conform to film classification designator 51-33-56d of L-F-336. The negative shall measure 11 by 14 inches (279 by 356mm) and shall have a neutral density of 1.00 ±0.05. It shall be free of scratches, kinks or other imperfections, and the curl shall not exceed 10 percent of the curved dimension when the negative is lying on a flat surface.

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4.6.4.3 Test paper. Paper used in the contact test shall conform to MIL-P-4672, type I, contrast 3. The paper, after exposure, shall be developed for a period of 1.5 minutes at 68 +2°F (20 ±1°C) in a developer conforming to MIL-D-4827. The stock solution of developer shall have been diluted by 2 parts of water to one part of stock solution.

4.6.4.4 Intensity of illumination. A print shall be made using photographic paper conforming to MIL-P-4672, type I, contrast 3 and a substantially neutral density negative of density 1.00 ±0.05. The paper shall be developed as prescribed in 4.6.4.3 and its reflective density determined with a suitable densitometer. The time required to produce a print shall be measured to determine compliance with 3.8.1. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6.4.5 Uniformity of illumination. A flash exposure shall be made using 11 by 14 inch (279 by 356mm) paper, conforming to MIL-P-4672, type I, contrast 3 and shall be exposed without use of a negative. After exposure, the paper shall be developed in the manner prescribed in 4.6.4.3 and the reflection density measured using a suitable densitometer at the lightest and darkest points to determine the maximum intensity differences of local irregularities in the entire field. The exposure time shall be chosen to give reflection densities of not more than 1.40 and not less than 0.70. Within this range, with the specified paper, a change of 0.10 in density corresponds to a change of 11 percent in illumination. Nonconformance to 3.8.2 shall constitute failure of this test.

4.6.4.6 Contact test. Three prints shall be made from the negative halftone screen, using the photographic paper and development prescribed in 4.6.4.3. One print shall be given a 1-second exposure, one print a 3-second exposure and one print a 5-second exposure. The negative halftone screen shall be centered on the printing area for uniformity of contact. Nonconformance to 3.8.3 shall constitute failure of this test.

4.6.5 Electromagnetic interference. The printer shall be tested for electromagnetic interference characteristics in accordance with MIL-STD-462. Nonconformance to 3.9 shall constitute failure of this test.

4.6.6 Interchangeability. The dimensions listed herein shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.11. When a listed dimension is not within specified limits, it shall be considered a major defect.

4.6.7 Visual and mechanical. The equipment shall be examined for the defects listed in MIL-STD-252. The equipment shall be visually inspected to determine that the requirements of 3.14 are met.

4.6.8 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.6.8.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

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4.6.8.2 Quality conformance inspection of packaging.

4.6.8.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.6.8.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.6.8.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table IV, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE IV. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight and measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G00478.

6. NOTES

6.1 Intended use. The printer is capable of either single or continuous cycle operation and is used to make contact prints from negatives up to 11 by 14 inches (279 by 356mm) in size in production type operations. It is commercial equipment with minimum modification, if necessary, to meet requirements of this specification.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2).
c. Group C failure action (see 4.5.3.3).
d. Level A or B preservation and packing (see section 5).
e. When first article inspection rough handling tests are not required.
f. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, approval, and the number of first article units. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examination test, and approval of first article.

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PH-649W/U. The contractor should apply for nomenclature in accordance with the applicable provision in the contract (see 1.1).

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of the corrective action (see 4.5.3.3).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.7 Power cable (see 3.5.3). CO-03-MGF (3/16) 0360 cable is available as National Stock No. (NSN) 6145-00-504-1015. When the quantity of this cable required for the contract is small, it may be uneconomical for the contractor to purchase it from vendors. If this occurs, the Government should furnish the cable from depot supply, with a corresponding reduction in contract cost.

6.8 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.9 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodian:
Army - ER
Navy - AS

Preparing activity:
Army - ER
(Project 6740-0141)

Review activity:
DSA-GS

User activities:
Army - MI
Navy - MC

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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